MACHINE WHITE PAPERS

Samsung SL-65A/3200 — Big Thicket Express

Large-envelope CNC lathe with steady-rest support for precision turning of long, heavy structural round parts. 22" OD × 119" L max turning (with steady rest).

Big Thicket Express · large-envelope heavy-duty CNC lathe with steady rest

This is the heaviest machine on the floor. When a customer sends a 20-inch forged blank and wants it turned to a finished OD with concentric internal features across ten feet of length, Big Thicket Express is the answer. The included steady rest is what makes long slender work possible — without it, deflection eats your tolerance.

What Big Thicket Express is for

Big Thicket Express is the largest turning envelope in the shop — 22" OD × 119" L with steady-rest support — and the steady rest is doing as much work as the swing. Length without support is a chatter problem, not a capacity one. Being able to hold a long heavy part in the middle while turning it is what makes a 22-inch diameter at ten feet a real working spec rather than a number on a page. When a part is both big and long, this is the only machine here that takes it.

Pick this machine when

- The part is over 16" in diameter — past the Mega lathes' ceiling.
- It is long AND heavy, and needs intermediate support to be turned without chatter.

- Large forgings, bar stock, and tubing in oilfield and structural alloys.
- You've been told a part is too big by shops that stopped at 16 or 18 inches.

> **When it's the wrong machine**
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> Small or medium work of any kind — the setup overhead on a machine this size is real, and a mid-size part will be quicker and cheaper on a Hi-Eco 35 cell. It also has no 10-inch through-hole; long tubular parts needing feed-through belong on Texaco Thunder even when the diameter would fit here.

From the floor

Steady-rest work is a setup discipline more than a machining one. The rest has to be placed on a turned, round, concentric journal — clamping onto as-received stock transfers that stock's runout straight into the finished part. So the sequence is always: establish a true journal, set the rest to it, then turn the rest of the part. Skipping that step is the most common reason long-part work comes out oval.

Pick your industry

Samsung SL-65A for Oil & Gas

Full spec sheet, typical oil & gas parts, alloys, tolerance expectations, documentation approach.

Samsung SL-65A for Aerospace

Full spec sheet, typical aerospace parts, alloys, tolerance expectations, documentation approach.

Samsung SL-65A for Military & Defense

Full spec sheet, typical defense parts, alloys, tolerance expectations, documentation approach.

Samsung SL-65A for Industrial & General Manufacturing

Full spec sheet, typical industrial parts, alloys, tolerance expectations, documentation approach.

Have a project?